AI-Driven High School Data Intelligence Platform

Linking, Classifying, and Auditing U.S. High Schools for College Admissions Context

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Prepared for Bob Henkins
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Summer 2026

Abstract

College admissions offices lack reliable, standardized information about the high schools their applicants attend, a gap that the College Board's own contextualization tool, Landscape, closed only partially before being discontinued in September 2025 for reasons unrelated to whether it worked. This paper develops and grounds a replacement: an internally owned, publicly sourced, auditable database and analytical platform that links, classifies, and contextualizes United States high schools for college admissions review. As of this writing, the platform consolidates ten public data sources, government census data, the National Center for Education Statistics, the Civil Rights Data Collection, College Board AP data, the International Baccalaureate Organization, and state and local sources, into a single master database of 25,577 United States high schools, with the National Assessment of Educational Progress, the College Scorecard, and the Office of Postsecondary Education identifiers planned as near-term additions. The pipeline links schools across identifier systems (NCES school IDs and CEEB codes, with OPE identifiers planned) using fuzzy matching, classifies each school on a five-tier curriculum rigor scale, analyzes per-student funding, benchmarks standardized test performance, and applies clustering methods to surface patterns among schools, all inside a Docker-based pipeline designed to run without engineering support once the team hands the project to the client. The paper evaluates these design choices against four bodies of scholarship: the construct and predictive validity of academic rigor measures, prior institutional attempts to deliver school context to admissions offices, the measurement literature on composite school-quality indicators, and the record-linkage and reproducibility methods the pipeline already implements. Three findings revise assumptions made in the project's own plan. First, the Civil Rights Data Collection, a federal census of nearly every public school, already publishes school-level AP, IB, and dual-enrollment participation, reducing the project's dependence on College Board institutional data access. Second, the College Board's September 2025 discontinuation of its Landscape tool, the closest existing analogue to this platform, was driven by post-*SFFA v. Harvard* legal and political pressure rather than by a commercial decision, which means the project inherits a governance and framing risk that its own plan currently asserts does not exist. Third, the federal statistical infrastructure the first two findings depend on is itself fragile: the National Center for Education Statistics was reduced to a handful of staff in 2025, and the office that runs the Civil Rights Data Collection is being relocated to the Department of Justice and is running roughly six months behind its own publication schedule as of this writing, so neither source should be treated as a permanently stable substitute for College Board data. The review further identifies an unresolved methodological tension: the rigor tier, built substantially from course-offering counts, will tend to reward resourced, high-socioeconomic-status schools, which runs counter to the platform's stated goal of surfacing under-served schools for outreach. Building on that literature, the paper then lays out the analytical methodology the team will implement on top of the master database itself: rigor classification with reported nominal and effective weights, a descriptive (not causal) funding overlay, feature engineering and dimensionality reduction ahead of model training, K-means and hierarchical clustering with explicit validation, standardized test benchmarking, and a train, test, and validation split with hyperparameter tuning and overfitting checks. A closing section of the literature review identifies five source gaps that this analysis will need to close: the predictive-validity literature on SAT and ACT scores, cluster-validation methodology, automated web-scraping and natural-language extraction for structured data acquisition, the coverage limits of NAEP and College Scorecard as outcome sources, and the missing-data literature relevant to administrative

education records. Data collection, exploratory analysis, and model training are in progress at the time of writing; results, discussion, and directions for future work are reported where available and otherwise marked as pending.

Contents

1 Introduction

College admissions offices have never had reliable, standardized information about the high schools their applicants attend. A student's transcript arrives without the context needed to interpret it: how many AP or IB courses that school even offers, how its per-student funding compares to its peers, or how its average SAT and ACT scores sit relative to similar schools regionally or nationally. Bastedo et al. (2023) document the scale of the problem directly, finding that admissions officers lack any high school context on roughly a quarter of applications, with public and majority-low-income schools least likely to supply it. For over a decade, the College Board's Environmental Context Dashboard and its successor, Landscape, were the closest thing to a standardized answer to this problem. As of September 2025 neither exists, and, as Section 2.3 details, its discontinuation had nothing to do with whether it worked.

Problem statement. No existing tool gives college admissions offices standardized, auditable high school context that is simultaneously honest about measuring curricular opportunity rather than outcome, decomposable down to the data behind every score rather than delivered as an opaque number, actively checked against reproducing the socioeconomic ordering of American schools, and owned by the institution that depends on it rather than licensed from a vendor whose product can be withdrawn for reasons unrelated to its design. Landscape came closer to this than anything else available, and it is gone. This paper develops and grounds a platform intended to close that gap for one institution, Northwestern University's Undergraduate Admissions and Enrollment office, in a form that is transparent and reproducible enough to survive the same pressures that ended Landscape. Section 2.9 develops this claim in full once the supporting literature has been reviewed.

Northwestern University's Undergraduate Admissions and Enrollment office, represented by Bob Henkins as project client, commissioned this capstone project to build that platform: an internally maintained, publicly sourced, reproducible database of United States high schools. The platform under construction consolidates data from ten public sources currently joined into the master database, including the U.S. Census Bureau's poverty and community-context data (U.S. Census Bureau, 2026b), the National Center for Education Statistics (NCES) school directory

(National Center for Education Statistics, 2026b), the Civil Rights Data Collection (CRDC) (U.S. Department of Education, Office for Civil Rights, 2026), the College Board's AP course and exam data (College Board, 2026), the International Baccalaureate Organization's school directory (International Baccalaureate Organization, 2026), and state and district sources including the Illinois State Board of Education's Report Card data (Illinois State Board of Education, 2025) and the Chicago Public Schools Opportunity Index (Chicago Public Schools, 2024), with the National Assessment of Educational Progress (NAEP) (National Center for Education Statistics, 2026a), the U.S. Department of Education's College Scorecard (U.S. Department of Education, 2026), and the Office of Postsecondary Education (OPE) ID database (U.S. Department of Education, Office of Postsecondary Education, 2026) planned as near-term additions, into a single relational master database, described in full in Section 3. A junction mapping system links each school across three identifier systems, NCES school IDs, CEEB codes, and, as a secondary goal, OPE identifiers, using a fuzzy-matching pipeline built on `rapidfuzz`. On top of that linked database, the platform classifies each school on a five-tier academic rigor scale (Below Average, Average, Demanding, Very Demanding, Most Demanding), analyzes per-student funding from Census F-33 and SAIPE data, benchmarks SAT and ACT performance across schools, and applies clustering methods to group institutions by location, academic profile, and funding. The entire pipeline runs inside Docker so that the client's team can re-run it annually without engineering support after the capstone team hands the project off.

This paper serves four purposes. Section 2 is a literature review that tests the project's existing design choices, the rigor construct, the record-linkage decision rule, the composite-indicator methodology, and the reproducibility strategy, against the relevant scholarship, identifying where the literature supports the current plan, where it pushes back, and where sources the project has not yet consulted would strengthen it, closing with the unified gap statement previewed above. Section 3 describes the data architecture built to close that gap: the eight source datasets, the master database schema, and the identifier-linkage system. Section 4 turns from what the literature says to what the team will actually do on top of that data: how the rigor tiers will be constructed and validated, how features will be engineered and reduced ahead of modeling, how clustering will be run and checked for validity rather than taken on faith, and how the resulting models will be trained, tuned, and evaluated for overfitting before anything is presented to the client. Sections 5 through 8 report results where they exist, discuss them, and lay out what remains, since, consistent with the capstone's own schedule, this document will be revised and extended through the term as the underlying analysis is completed.

2 Literature Review

2.1 A Note on Three Findings That Revise the Project Plan

Three findings in this review revise the project's own plan: the first two contradict claims the plan makes outright, and the third attaches a currency risk to the first that the plan does not yet acknowledge.

First, the plan states that no public source publishes school-level AP participation. This is not correct: the Civil Rights Data Collection (CRDC), a mandatory biennial census of every public school and district receiving federal funds, publishes school-level AP course counts and enrollment, IB Diploma Programme enrollment, dual-enrollment participation, SAT/ACT participation, and advanced math and science offerings, keyed to NCES identifiers with no fuzzy matching required (U.S. Department of Education, Office for Civil Rights, 2026). It does not publish AP exam scores, so the coursework-versus-performance distinction developed below still binds, but CRDC substantially reduces the project's dependency on a College Board institutional-access request and may make the team's scraped IB file redundant, or independently verifiable.

Second, the College Board discontinued Landscape, the closest existing analogue to this platform, in September 2025. Its withdrawal is almost certainly the vendor-discontinued-data-delivery event the project plan's business case describes, but it was withdrawn for post-*SFFA v. Harvard* legal and political reasons, not commercial ones (Section 2.3), which means the plan's assertion that no government approval or socio-cultural considerations apply is difficult to sustain.

Third, the federal infrastructure the first two findings both depend on is itself under strain. NCES, which maintains the Common Core of Data this project treats as its authoritative directory (National Center for Education Statistics, 2026b), was reduced to five employees in a March 2025 reduction in force, and the Institute of Education Sciences that houses it lost over one hundred more staff in the same round of cuts, after roughly $900 million in IES contracts were canceled the month before (Palmer, 2025). The Department confirmed it still maintained NAEP, College Scorecard, and IPEDS contracts at the time, but the CCD's own future maintenance schedule, and with it the "annually re-runnable pipeline" premise the client hand-off depends on, is no longer something the plan can simply assume. Separately, the Office for Civil Rights, which runs CRDC, is being relocated to the Department of Justice, and as of July 2026 its 2023-24 release is running roughly six months behind schedule with no public explanation (Mehta, 2026). This does not undermine the first finding, CRDC still intends to publish the data, but it means the project should not treat CRDC as a dependable substitute for College Board access without a fallback (state-level sources such as ISBE, or direct school outreach) if the release does not arrive in time.

Given these three findings, the review below asks not whether the project's approach has been tried before, but what four literatures, academic rigor's construct validity, prior institutional attempts at school context, composite-indicator measurement, and record linkage and reproducibility, say about the specific design choices already made, since the pipeline is largely built and the value of a review at this stage is identifying which choices are supported, which are under-supported, and which are contradicted.

2.2 Academic Rigor: Construct and Predictive Validity

2.2.1 The foundational claim, and its limits

Adelman's two U.S. Department of Education longitudinal studies provide the project's strongest empirical support. *Answers in the Tool Box* (Adelman, 1999) followed the High School and Beyond cohort and found that curriculum intensity and quality was the strongest predictor of bachelor's degree completion, ahead of test scores and GPA, accounting for roughly 43 percent of variance.

The Toolbox Revisited (Adelman, 2006) replicated this on NELS:88/2000 and strengthened it: intensity mattered more, senior-year test scores mattered less, and students in the top quintile of a 31-level curriculum-intensity measure earned degrees at a 95 percent rate. Adelman's work justifies building a rigor variable from curriculum offerings rather than test scores, but it measured individual transcripts, not school characteristics; inferring a student's curricular intensity from their school's course catalog is an ecological inference the project should name as such rather than treat as validated at the school level.

2.2.2 Offerings versus performance, and what that means for the tier

Geiser and Santelices (2004) found that AP and honors course-taking had little predictive validity for UC admissions outcomes once other factors were controlled, while AP exam performance was a strong predictor; the College Board's response (Camara and Michaelides, 2005) disputed the framing but not the core result. The literature has largely converged: exam performance carries signal, coursework enrollment alone does not. This matters directly for the rigor model as specified, which is built primarily from availability data, IB flags (with no U.S. exam scores), state AP aggregates, and CRDC's AP course counts, exactly the variable this literature finds weakest. The tiers will therefore measure opportunity structure, what a school offers, not academic outcome, and should be named accordingly ("Most Opportunity-Rich" rather than "Most Demanding").

2.2.3 Availability is confounded with poverty and size, and IB evidence is more favorable but still selection-prone

Kolluri's review (Kolluri, 2018) documents that low-income students enroll in AP, where offered, at under a third of the rate of higher-income peers, and nearly all quantitative AP-effectiveness studies are correlational. The GAO (U.S. Government Accountability Office, 2018) found directly that high-poverty, smaller schools offer fewer college-prep courses. A rigor tier built from offering counts will therefore substantially proxy school size and affluence, meaning the project's funding and poverty overlay is not an independent enrichment layer but collinear with the rigor outcome itself; any feature-selection step should report that correlation explicitly. IB evidence is more favorable: Coca et al. (2012), studying CPS IBDP graduates against a matched comparison group, found IBDP students roughly 40 percent more likely to attend four-year colleges, with significantly higher persistence, a well-suited citation for the project's CPS pilot region (Chicago Public Schools, 2024). But IB students are self-selected, and IB-published outcome research typically compares them to national averages rather than matched controls, so Coca and colleagues should be cited in preference to IB's own studies.

2.3 Prior Art: Delivering School Context to Admissions Offices

2.3.1 The evidence for the platform's existence

Bastedo and colleagues provide the most rigorous evidence that school-context data changes admissions behavior. In field experiments with admissions officers at eight universities, officers practicing holistic review were more likely to recommend admitting low-socioeconomic-status applicants when given contextual data, with priming effects persisting even for readers not shown the dashboard (Bastedo et al., 2022). Bastedo et al. (2023) extended this to outcomes, finding that contextualized high school GPA predicted college success more strongly than contextualized test scores, and documented the gap this project is positioned to fill: roughly a quarter of applications arrive with no standardized high school context, worst for public and majority-low-income schools. This is a stronger justification for the project than the vendor-loss framing in the current business case: context data measurably improves both equity and prediction.

2.3.2 The cautionary history

The College Board's own attempt at this problem is best read as three attempts, not one static tool. The first, the Environmental Context Dashboard (piloted 2016, publicized 2019), reduced school and neighborhood indicators to a single 1-to-100 "adversity score"; widespread criticism that a single number could not represent a student's circumstances forced its withdrawal within months.

The second, Landscape, corrected exactly that flaw by presenting the same indicators separately with no composite score. It worked on its own terms: it ran for six years and, per Mabel et al. (2022), modestly increased offers to students from high-challenge schools, though with little effect on enrollment absent financial-aid adjustments; Mabel and colleagues also found socioeconomic proxies to be a poor, if correlated, substitute for race. Landscape's end was a third attempt, this time external: the College Board discontinued it in September 2025 citing evolving federal policy on demographic and geographic information, in the wake of *SFFA v. Harvard*, an executive order on "hidden racial proxies," a DOJ memo on geographic recruiting, and direct pressure from Students for Fair Admissions. Commentators across the political spectrum, including an SFFA expert witness himself, called the decision unfortunate, since Landscape was race-neutral and several *SFFA* justices had explicitly endorsed socioeconomic factors as a permissible diversity route (Philogene, 2025). Mabel, in the same reporting, called it "a politically savvy move to avoid risk," not a response to any demonstrated flaw (Philogene, 2025). One attempt was withdrawn for a measurement flaw and corrected; one worked and was studied; one was killed for reasons unrelated to whether it worked. That third outcome is why this project's literature-grounded, auditable design does not by itself close the gap Landscape leaves.

2.3.3 What this means for the project

Three consequences follow. First, the 2019 backlash was specifically about collapsing multidimensional context into one number; this project's five-tier rigor classification is a single number by another name, so its constituent indicators must remain visible and decomposable, extending the project's existing match-tier auditability commitment to the rigor score itself. Second, the project's risk register is missing a risk: it currently states that no government, socio-cultural, or gender-related considerations apply, but Landscape's history shows that a race-neutral, publicly sourced school-context tool attracted sustained legal and political attention regardless, and the platform's gap-detection deliverable, identifying schools for expanded outreach, is functionally geographic recruiting, the practice named in the DOJ memo; this belongs in front of the client and

Northwestern's counsel, not asserted away. Third, with Landscape gone, institutions must build their own contextualization or go without, which is as close as this literature comes to a direct argument for why an in-house, transparent platform is worth building (Philogene, 2025).

2.4 Measuring School Quality: What the Composite-Indicator Literature Warns

2.4.1 Rankings built on levels measure demographics, not quality

The Stanford Education Data Archive, assembling roughly 350 million standardized test scores onto a common scale for grades 3 through 8, is the closest existing methodological analogue to this project (Ho, 2020). Reardon's core SEDA finding (Jang and Reardon, 2019) is that average achievement levels are powerfully predicted by district socioeconomic composition, while academic growth from grade 3 to 8 bears little relationship to early-childhood advantage; growth, not levels, is "a much better measure of school quality," and levels-type data "should not be used to rank school districts whose performance differs only slightly." A five-tier ranking built from levels-type indicators (AP offerings, assessment levels, funding) will substantially reproduce the socioeconomic ordering of American high schools, ranking affluent suburban schools highest and under-resourced schools lowest, the opposite of what the gap-detection deliverable is for. SEDA itself covers grades 3 through 8 only, so it cannot supply high-school achievement directly, but it can supply useful feeder-district context.

2.4.2 Weights, effective weights, and the funding overlay

When a composite is built from correlated indicators, nominal weights diverge from effective weights, the actual influence each indicator exerts given its variance and covariance with the others; a 2024 demonstration on Colorado's School Performance Framework shows this divergence can materially undermine interpretation (Center for Assessment, Design, Research and Evaluation, 2024). Because this project's rigor features (AP counts, IB flags, funding, poverty) are strongly intercorrelated, effective weights should be computed and reported alongside nominal ones, and tier robustness should be tested against alternative weightings, a cheap addition that would substantially strengthen the "objective and reproducible" claim made to the client. On funding specifically, Jackson, Johnson, and Persico provide the strongest causal evidence that spending matters (Jackson et al., 2016): using court-mandated finance reforms as exogenous shocks, they find a 10 percent sustained increase in per-pupil spending raises completed education by 0.27 years and wages by 7.25 percent, with larger effects for low-income children. But their result concerns spending changes over time, not cross-sectional spending levels, exactly the confounded comparison this project's district-grain, LEAID-joined data represents; the funding overlay is worth having as descriptive context, not as a causal input to rigor.

2.5 Record Linkage Without a Shared Identifier

2.5.1 The theory the pipeline already implements

Fellegi and Sunter formalized probabilistic record linkage (Fellegi and Sunter, 1969), building on Newcombe's earlier vital-records matching (Newcombe et al., 1959): compare record pairs on multiple quasi-identifiers, compute a likelihood ratio, and apply two thresholds producing a three-way decision (link, possible link routed to clerical review, non-link) that provably minimizes the indeterminate middle set for specified error rates. The project's auto-accept, review, and reject tiering is a Fellegi-Sunter decision rule, which should be stated explicitly, since it converts what currently reads as an engineering convenience into a design grounded in over fifty years of statistical theory. It also identifies a limitation worth naming: Fellegi-Sunter's optimality assumes conditional independence among comparison attributes, an assumption the classical algorithm shares with naive Bayes and one known to be violated in practice. If time permits, Enamorado et al. (2019)'s EM-based implementation and Christen (2012)'s standard reference text point toward a properly probabilistic extension that would let the team report calibrated match probabilities rather than uncalibrated similarity scores.

2.5.2 Token ratios, and where they fail

Cohen et al. (2003) found that token-based and hybrid string-comparison methods outperform pure edit distance for entity names, validating the choice of `rapidfuzz` token ratios over Levenshtein distance for school names, where reordering and abbreviation dominate. It also predicts the failure the team observed: `token_set_ratio` scores a short name that is a strict subset of a longer one as a perfect match ("Academy High" scoring 100 against a much longer name while collapsing to roughly 36 on `token_sort_ratio`), which the team's two-signal guard correctly, if ad hoc, corrects for. The literature also suggests a cheap improvement: TF-IDF down-weighting of high-frequency, low-information tokens ("High," "School," "Academy"), the change most likely to widen the auto-accept tier without admitting false positives, relevant given that only two of seventy-seven Illinois private schools in the ISBE data (Illinois State Board of Education, 2025) currently clear the bar.

2.5.3 The crosswalk problem is known and unsolved

The project is not the first to hit this. The University of Colorado Boulder's `ceeb_nces_crosswalk` (University of Colorado Boulder, Office of Data Analytics, 2026) uses a three-stage pipeline (an existing Davenport crosswalk, fuzzy matching, and Mechanical Turk resolution with redundant coders) to cover 21,592 matched schools. The project's 48 percent auto-accept rate against this reference is therefore not anomalous, since even the reference is a partial, crowd-augmented artifact; match rates against it should be described as agreement, not accuracy. CU Boulder's redundant-coder design is worth proposing for the review tier, and the repository accepts contributions, so publishing this project's validated matches back upstream would be a concrete, zero-cost contribution.

2.6 Reproducibility and Pipeline Engineering

The containerization decision has direct support. Boettiger (2015) catalogues why computational work fails to reproduce (dependency drift, undocumented environment state, code rot) and argues Docker addresses this through OS-level virtualization and versioned, human-readable Dockerfiles;

Wiebels and Moreau (2021) extend this to a practical tutorial, noting containers support reproducibility during a project, by keeping collaborators' environments identical, not only after it. This maps directly onto the project's stated need: a pipeline the client's team can re-run annually without engineering support. The one gap is that Boettiger's argument is about environment reproducibility, while the project's stated risk of data-source schema instability is data reproducibility, a distinct problem Docker does not solve; schema-aware validation and pinned source-year snapshots are the correct response, and the paper should keep the two claims separate.

2.7 Synthesis: What the Literature Supports, and Where It Pushes Back

The design is well supported where curriculum intensity is used as the rigor construct (Adelman, 1999, 2006), where school-context data is treated as an equity- and accuracy-improving intervention (Bastedo et al., 2022, 2023), where the three-tier linkage decision follows Fellegi-Sunter (Fellegi and Sunter, 1969), where token-based matching is chosen over edit distance (Cohen et al., 2003), and where containerization is used for a non-engineer-maintained pipeline (Boettiger, 2015); per-pupil spending is also a variable with genuine causal purchase (Jackson et al., 2016).

The literature pushes back in four places. Course availability is a weak outcome predictor while exam performance is strong (Geiser and Santelices, 2004), and the rigor model as specified is built mostly on the former. Offering counts are confounded with school size and poverty (U.S. Government Accountability Office, 2018; Kolluri, 2018), making the funding overlay collinear with the rigor target rather than independent of it. Levels-based composites reproduce socioeconomic ordering and should not rank schools that differ only slightly (Jang and Reardon, 2019; Ho, 2020), in direct tension with the gap-detection deliverable. And a single composite score has already been tried and withdrawn once (2019) and killed again under legal and political pressure the second time (2025), which does not support the plan's assertion that no socio-cultural or governmental considerations apply.

Unused sources worth adding include CRDC for AP, IB, dual-enrollment, and SAT/ACT participation (U.S. Department of Education, Office for Civil Rights, 2026), SEDA for feeder-district context, TF-IDF weighting as a matcher improvement, and the CU Boulder crosswalk (University of Colorado Boulder, Office of Data Analytics, 2026) as both a validation reference and a contribution target.

2.8 Remaining Gaps: Sources Not Yet Addressed

Comparing this review against the full capstone scope, data integration, rigor classification, funding analysis, performance benchmarking, and clustering, surfaces five gaps not covered by the synthesis above.

The first concerns standardized test performance. The project's benchmarking objective compares SAT and ACT scores across schools, yet the SEDA discussion above already shows achievement-level indicators reproduce socioeconomic ordering, and there is every reason to expect the same of SAT and ACT scores. Two sources belong in this review rather than sitting uncited. Geiser and Studley, examining 77,893 UC applicants across three admission cohorts (1996 to 1999), found SAT I more sensitive to socioeconomic status than SAT II or GPA, with SAT II and GPA consistently better predictors of UC success and SAT I contributing almost no incremental validity (Geiser and Studley, 2002). Bettinger, Evans, and Pope found only the English and Math ACT subtests highly predictive of college outcomes, with Science and Reading adding little or nothing, a result robust across samples and specifications (Bettinger et al., 2013). Together these say that a peer-comparison feature built from an aggregate, undisaggregated SAT or ACT score will average components with very different predictive validity and very different sensitivity to family income; benchmarking should report section-level aggregates where the data permit it. Coaching effects and the test-optional literature remain unaddressed.

The second concerns clustering, and is narrower than it looks once the project's own methodology is accounted for. Section 4 already selects cluster count via the gap statistic (Tibshirani et al., 2001), cross-checked against silhouette scores. What this review lacks is validation of that method's behavior on data with this project's specific confounding structure, offerings, funding, and poverty already known to load on one socioeconomic factor, and any null-result precedent for what a weak validation outcome looks like on school-level administrative data. A previous project on this team found near-zero silhouette scores can reveal that the data, not the method, is the limiting factor; the same possibility should be anticipated here.

The third concerns the project's own framing as AI-driven: the stated approach includes web scraping and NLP-based extraction for unstructured data, yet this review contains no literature on the reliability of such extraction from educational sources, and at minimum a citation establishing typical LLM or NLP extraction error rates is needed before the team can set an honest accuracy target.

The fourth concerns two named sources this review does not discuss: NAEP (National Center for Education Statistics, 2026a), whose coverage and suitability for school-level inference is unevaluated here, and the College Scorecard (U.S. Department of Education, 2026), the most direct available measure of college-going outcomes, whose restriction to federal aid recipients and cohort-based reporting lag are not addressed by any source currently cited for college-going outcomes.

The fifth concerns OPE-identifier linkage. Section 2.5 develops the NCES-CEEB crosswalk problem at length, but whether OPE identifiers present the same many-to-many ambiguity, or a different one, is unknown; the Fellegi-Sunter framework should be extended to this identifier system explicitly. A related, smaller gap is missing-data handling: the project names processing incomplete records as a core feature but this review does not engage the missing-data literature (the distinction between data missing completely at random, at random, and not at random, and its implications for imputation) that bears on how gaps should be treated rather than simply dropped.

These five gaps should be closed, or explicitly scoped as out of bounds, before the corresponding features move from plan to implementation in Section 4.

2.9 The Gap This Platform Fills

Figure 1 lays out the argument of this section before making it in prose: four literatures, reviewed separately above, converge on one gap.

Bastedo and colleagues establish that admissions offices need standardized context and lack

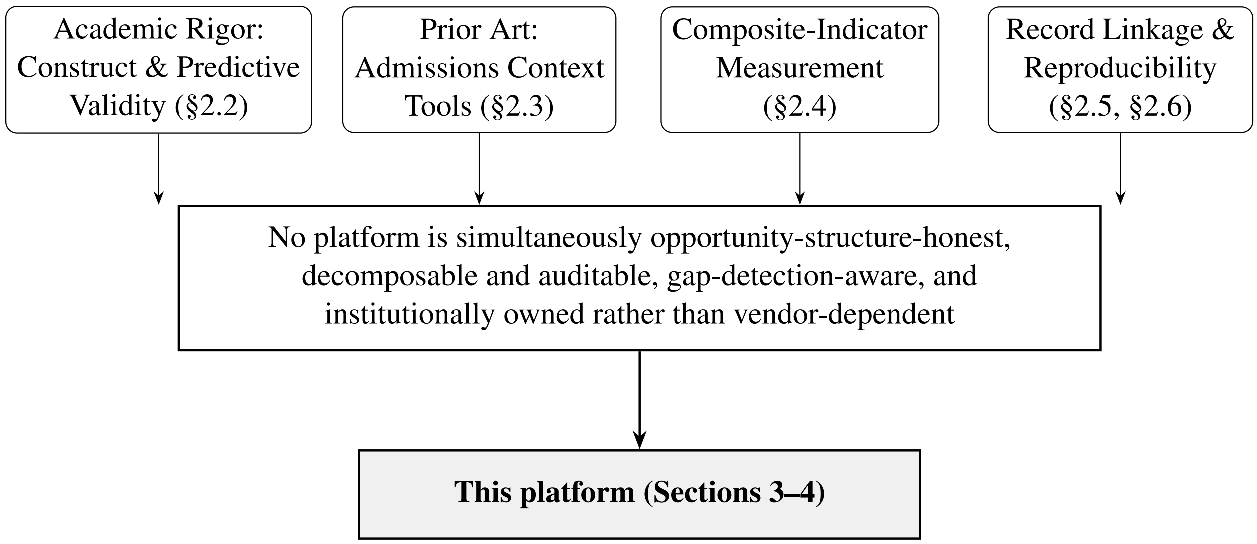

Figure 1: The four literatures reviewed in Section 2 converge on a single unresolved gap: no existing tool is simultaneously honest about measuring opportunity rather than outcome, auditable down to its component features, actively checked against reproducing socioeconomic ordering, and owned by the institution that depends on it rather than licensed from a vendor. Sections 3 and 4 describe the platform built to close it.

it for roughly a quarter of applicants (Bastedo et al., 2023). The College Board's three attempts (Section 2.3) show the measurement problem was solvable, and was solved when the single adversity score became Landscape's separate indicators, but the resulting tool was still killed for reasons external to its design. Separately, the rigor and composite-indicator literatures (Sections 2.2, 2.4) show that any tool built primarily from offerings and levels-type indicators will tend to reproduce the socioeconomic ordering of American high schools, precisely what a platform meant to surface under-served schools should avoid, defensible only if the tool is transparent enough for a reader to see and correct for it.

No source in this review describes a platform that is simultaneously honest about measuring opportunity rather than outcome; decomposable and auditable down to its underlying features; actively checked against reproducing socioeconomic ordering; and owned by the institution that uses it rather than licensed from a vendor whose product can be discontinued for reasons unrelated to whether it works. Landscape came closest and was still not that platform, not because its measurement design failed, but because it was not owned by the institutions that depended on it.

That is the gap this platform is built to fill, and it is why the analytical choices in Section 4, naming the rigor tier honestly, reporting nominal against effective weights, validating clusters rather than assuming them, keeping every tier decomposable, are this literature review's argument carried into the pipeline.

3 Data

Section 2.9 argued that closing the gap left by Landscape requires a platform that is owned by the institution that depends on it and built entirely from public data. This section describes the data architecture that makes that possible: what is consolidated, from where, and how it is linked. Section 4 then describes what is done with it.

3.1 Source Inventory

As of this writing, the master database consolidates ten source groups into 25,577 linked school records, 24,223 public and 1,354 private, spanning all 50 states and the District of Columbia. NCES supplies the primary school directory, for public schools through the Common Core of Data and for private schools through the Private School Survey (National Center for Education Statistics, 2026b). EDFacts supplies four-year adjusted-cohort graduation rates and statewide assessment-participation rates for roughly 70 percent of public schools, drawn from the COVID-affected 2020-21 school year, a limitation that should be read into any use of those two variables. The Civil Rights Data Collection supplies school-level AP, IB, dual-enrollment, and SAT/ACT participation, along with counselor staffing and teacher-certification rates, for roughly 78 percent of public schools (U.S. Department of Education, Office for Civil Rights, 2026); this is the source Finding One in Section 2 identifies as reducing the project's dependence on a College Board institutional-access request. The College Board's own AP course and exam participation data (College Board, 2026) and the International Baccalaureate Organization's school directory (International Baccalaureate Organization, 2026) supply the remaining opportunity-structure indicators; the IB directory is currently fuzzy-matched against every private school, with none yet confirmed at the auto-accept tier (Section 2.5), so IB participation is not yet a usable input. The U.S. Census Bureau supplies county-level income, education, and poverty context through the American Community Survey and SAIPE (U.S. Census Bureau, 2026b), covering 94 percent of schools. Per-pupil finance data from the Annual Survey of School System Finances (F-33) (U.S. Census Bureau, 2026a) is planned but not yet joined at national scale; the Illinois State Board of Education's Report Card data (Illinois State Board of Education, 2025) currently stands in for it, covering roughly 2.6 percent of schools, alongside the Chicago Public Schools Opportunity Index (Chicago Public Schools, 2024) supporting the project's CPS pilot region. Three sources named in the project's original plan, NAEP (National Center for Education Statistics, 2026a), the College Scorecard (U.S. Department of Education, 2026), and the OPE ID system (U.S. Department of Education, Office of Postsecondary Education, 2026), remain planned additions and are not yet reflected in the current file (Section 8).

3.2 Master Database and Identifier Linkage

The master database is a PostgreSQL 16 instance, built and re-buildable through Docker and `docker compose`, that consolidates the source datasets above through an ETL pipeline written in pandas and SQLAlchemy, with bulk loads handled by psycopg2's native `COPY` for performance. NCES (National Center for Education Statistics, 2026b) serves as the primary school-identity source, with CEEB codes drawn from College Board data (College Board, 2026) joined in through a fuzzy-matching layer built on `rapidfuzz` token ratios; OPE identifiers (U.S. Department of Education, Office of Postsecondary Education, 2026) are a planned secondary join and are not yet incorporated. As of this writing, 75 percent of public schools and 61 percent of private schools carry a CEEB code: 16,187 public and 613 private records sit at the auto-accept tier, and 2,076 public and 208 private records sit in manual review, consistent with the Fellegi-Sunter decision rule described in Section 2.5. Figure 2 summarizes the resulting architecture, from source ingestion through the analytical components described in Section 4; CRDC has been added to the pipeline since this diagram was first drafted, and NAEP, the College Scorecard, and OPE ID, shown as planned inputs, are not yet joined.

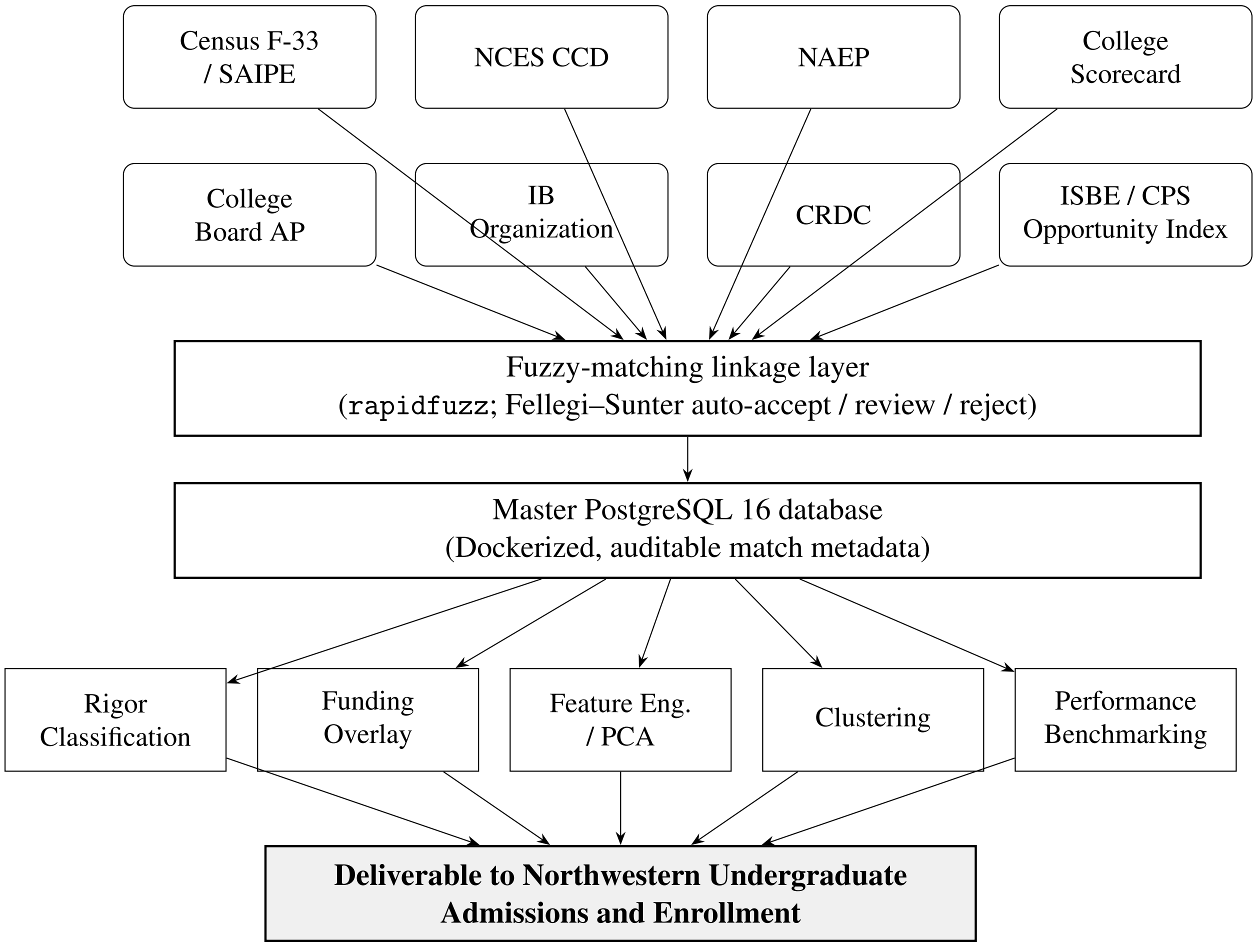

Figure 2: Data architecture: source groups are linked through a fuzzy-matching layer implementing a Fellegi–Sunter accept/review/reject decision rule (Section 2.5) into a single Dockerized master database, which feeds the five analytical components described in Section 4. NAEP, College Scorecard, and OPE ID are shown as planned inputs and are not yet joined into the current build.

Consistent with the Fellegi-Sunter framework discussed in Section 2.5, every match is assigned to one of three tiers, auto-accept, manual review, or reject, rather than to a single accept-or-reject decision, and every accepted match carries metadata documenting the similarity scores and token comparisons that produced it, so that a reviewer or the client can audit any individual linkage decision. The same tiering currently applies to the IB directory match: all 1,354 private-school candidates identified by the fuzzy matcher are split between manual review (588) and reject (766), with none yet confirmed at auto-accept, so IB participation is not yet a usable rigor-classifier input and should be gated on match tier rather than treated as available (Section 2.2). TF-IDF down-weighting of high-frequency tokens ("High," "School," "Academy") will be tested as a low-cost improvement to the current auto-accept rate ahead of a full probabilistic (EM-based) implementation, if time allows.

4 Methods

Section 2.9 named the gap in one sentence: no existing platform is simultaneously opportunity-structure-honest, auditable, gap-detection-aware, and institutionally owned. This section is the team's attempt to fill exactly that gap, translating the review's implications into a concrete methodology applied on top of the data architecture described in Section 3, rather than leaving them as commentary.

4.1 Academic Rigor Classification

Each school will be assigned to one of five ordinal rigor tiers (Below Average, Average, Demanding, Very Demanding, Most Demanding) built from AP course counts and enrollment (College Board, 2026), IB course counts and enrollment (International Baccalaureate Organization, 2026), CRDC's advanced-coursework indicators where adopted (U.S. Department of Education, Office for Civil Rights, 2026), and standardized test participation. Consistent with Section 2.2, the tier will be documented and described as a measure of curricular opportunity structure rather than of academic outcome or school quality, since it is built from offerings and enrollment rather than exam performance. Because the underlying features (AP and IB counts, per-pupil funding, poverty rate) are strongly intercorrelated, the team will compute and report both the nominal weights assigned during tier construction and the effective weights those features actually carry given their variances and covariances, following the demonstration in Center for Assessment, Design, Research and Evaluation (2024). A sensitivity analysis will vary the weighting scheme and report the resulting rank-order correlation and the number of schools that change tier, so that the "objective and reproducible" claim made to the client rests on evidence rather than assertion. The correlation between the poverty and funding overlay and the rigor score itself will be reported explicitly rather than presented as an independent enrichment layer.

The tier's feature set is itself contingent on data availability rather than fixed in advance. Which inputs a given school's tier is actually built from, College Board AP data if an institutional-access request is granted, CRDC's advanced-coursework counts if that release lands in time (Section 2's third finding notes this is not guaranteed given the Office for Civil Rights transition), or IB participation alone where neither is available, will vary school to school rather than being uniform across the database. Rather than silently imputing missing inputs or excluding schools with incomplete coverage, the team will log which features were actually available for each school's tier assignment and report source-coverage rates alongside the tier distribution. The sensitivity analysis described above will be extended to test how tier assignments shift under different source-availability scenarios, in particular a CRDC-available versus CRDC-unavailable comparison, so that a tier's reliability can be assessed independent of the specific mix of federal data that happened to be current when the pipeline was run.

4.2 Funding and Resource Overlay

Per-student funding will be computed from Census F-33, the Public Elementary-Secondary Education Finance Data (U.S. Census Bureau, 2026a), and Small Area Income and Poverty Estimates (SAIPE) (U.S. Census Bureau, 2026b), joined on LEAID at the district level and propagated to the school level with an explicit note that this join can produce fanout for large districts, a known issue the team has already encountered. Consistent with Jackson et al. (2016), the funding overlay will be presented as descriptive context rather than as a causal input to the rigor score, since Jackson, Johnson, and Persico's causal estimates concern spending changes identified through policy shocks, not cross-sectional spending levels of the kind this project observes.

4.3 Feature Engineering and Dimensionality Reduction

Following exploratory data analysis, the team will apply principal component analysis and related feature-selection techniques to the correlated rigor, funding, and poverty features identified above, both to reduce dimensionality ahead of clustering and to make the collinearity documented in Section 2.2 explicit and quantified rather than implicit. Any engineered feature will be logged with the raw inputs it was derived from, so that a rigor tier or cluster assignment remains traceable back to source data, consistent with the auditability standard set for the record-linkage pipeline.

4.4 Pattern Recognition and Clustering

The team will apply K-means and hierarchical clustering to group schools by location, academic profile, and funding characteristics, in order to surface patterns among schools independent of the ordinal rigor tier. Because Section 2.8 identifies cluster validity as a literature gap the current review does not close, the team will not treat a clustering result as self-validating: the number of clusters will be selected using the gap statistic (Tibshirani et al., 2001) and cross-checked against silhouette scores, and any clustering result will be checked against the possibility, flagged in Section 2.4, that recovered clusters simply reproduce the same socioeconomic ordering that the rigor tier already captures. If silhouette scores or gap-statistic results are weak, that will be reported as a finding about the data rather than suppressed or reframed as a modeling failure, consistent with the team's experience on a prior project where near-zero silhouette scores revealed that the data, not the clustering method, was the limiting factor.

4.5 Performance Benchmarking

SAT and ACT scores, where available at the school level, will be compared across schools and used to enable peer-group comparisons by region, funding tier, and rigor classification. Consistent with the gap identified in Section 2.8, this benchmarking will be presented descriptively rather than as a validated predictor of school quality until the predictive-validity and socioeconomic-confounding literature on SAT and ACT scores has been incorporated into the paper; the SEDA finding that achievement levels reproduce socioeconomic ordering (Section 2.4) is treated as a reason for caution here as well, not only for the rigor tier.

4.6 Validation Plan

Data will be split into training, testing, and validation sets before any model-fitting begins, consistent with the project's data-access and distribution plan. Model training (for the clustering and, if pursued, rigor-classification components) will be followed by a dedicated validation phase covering hyperparameter tuning and explicit checks for overfitting and underfitting, rather than treating a single training run as final. At the pipeline level, validation also includes the byte-for-byte, cross-machine reproducibility test and the row-count and schema-validation assertions described in Section 2.6, since data reproducibility and model validity are treated as distinct concerns that both need to pass before the client sign-off gate.

5 Results

This section is pending. Data collection, matching, exploratory data analysis, and feature engineering are actively in progress at the time of writing, and model training for the rigor-classification and clustering components described in Section 4 has not yet been completed. This section will report, at minimum: current auto-accept, review, and reject rates for the NCES-CEEB and NCES-OPE linkage tiers; the nominal and effective feature weights underlying the rigor classification, together with the sensitivity analysis described in Section 4; the number of clusters selected by the gap statistic and the corresponding silhouette scores, reported honestly whether or not they indicate strong structure; and the correlation between the funding and poverty overlay and the rigor score, as flagged in Section 2.2. It will be completed once these analyses are finished, consistent with the capstone's own schedule.

6 Discussion

This section is pending the results described in Section 5. Once available, the discussion will interpret those results against the two open questions this paper has already identified rather than treating them as self-evidently favorable: whether the rigor tier and the clustering results, once produced, in fact reproduce the socioeconomic ordering the literature in Section 2.4 predicts they will, and if so, what that means for the platform's gap-detection goal; and whether the sensitivity analysis on rigor-tier weights described in Section 4 shows the tier assignments to be robust enough to support the "objective and reproducible" claim made to the client. A weak or null clustering result, per the validation plan in Section 4, will be reported and discussed as a finding about the data rather than omitted.

7 Conclusions

This paper is a working document, consistent with the capstone's own schedule, which calls for the introduction, literature review, and methods sections to be drafted early and revised through the term as the underlying analysis is completed. At its current stage, the paper's contribution is threefold. First, it grounds the platform's central design choices, the rigor construct, the Fellegi-Sunter-style record-linkage decision rule, the composite-indicator methodology, and the containerized reproducibility strategy, in the relevant scholarship, and corrects three assumptions in the project's own plan: that no public source publishes school-level AP participation; that no government, socio-cultural, or governance considerations apply to the platform; and that the federal data sources the plan treats as stable, in particular NCES and the Civil Rights Data Collection, can be assumed to remain reliably staffed and released on schedule. Second, it identifies a methodological tension that the team has not yet resolved, that a rigor tier built substantially from course offerings will tend to reward the same resourced, high-socioeconomic-status schools that the platform's gap-detection goal is meant to look past, and it proposes that the clustering and validation plan in Section 4 treat this tension as an explicit finding rather than an incidental side effect. Third, it lays out the analytical methodology, from feature engineering and dimensionality reduction through clustering, performance benchmarking, and a formal train, test, and validation split, that the team will implement on top of the already-built master database over the remainder of the term.

What remains is to close the five source gaps identified in Section 2.8, complete the feature engineering and model training described in Section 4, and report results, including any weak or null clustering outcomes, honestly rather than only where they confirm the platform's premise. The team expects to revise this paper as those results come in, finalize it once client sign-off is secured, and submit it alongside the capstone's final presentation.

8 Directions for Future Work

Several directions are already identified and do not depend on the results in Section 5. Closing the five source gaps in Section 2.8, the predictive-validity literature on SAT and ACT scores, cluster-validation methodology, automated extraction reliability, NAEP and College Scorecard coverage limits, and the missing-data literature, should happen before the corresponding features are presented to the client as finished rather than descriptive. On the linkage side, extending the Fellegi-Sunter framework in Section 2.5 to OPE identifiers, testing TF-IDF token weighting to widen the auto-accept tier, and moving from the current heuristic tiering to a properly probabilistic, EM-based implementation (Enamorado et al., 2019) are all concrete, scoped next steps. Finally, publishing this project's validated NCES-CEEB matches back to the CU Boulder crosswalk repository (University of Colorado Boulder, Office of Data Analytics, 2026) remains a low-cost contribution to the field that the team can make independently of the client engagement's own timeline.

9 Data Availability

The source datasets underlying the master database are public and are cited individually in Section 3. The compiled and matched master database itself, including linkage metadata and any school-level rigor or cluster assignments, is being developed for Northwestern Undergraduate Admissions and

Enrollment as part of a client engagement and is not being publicly released at this time. Where feasible, the team intends to contribute validated NCES-CEEB match records back to the University of Colorado Boulder's public `ceeb_nces_crosswalk` repository (University of Colorado Boulder, Office of Data Analytics, 2026), independent of the client deliverable.

10 Code Availability

The ETL, linkage, and analysis pipeline is version-controlled and runs inside Docker via `docker compose up -build`, so that it can be rebuilt byte-for-byte across machines without engineering support, consistent with the reproducibility standard discussed in Section 2.6. The repository is currently private to the capstone team and the client. Whether any components, in particular the matcher improvements discussed in Section 2.5, are released publicly at the project's conclusion is a decision for the team and the client and has not yet been made.

11 Acknowledgements

The authors thank Bob Henkins and Northwestern Undergraduate Admissions and Enrollment for sponsoring this project and for the opportunity to surface the governance and framing risks discussed in Section 2.3. The authors also acknowledge the Northwestern University Master of Science in Data Science program, and the writing-process framework developed for this capstone, which shaped the structure and revision process of this paper.

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A Glossary of Identifiers and Acronyms

Acronym	Meaning
NCES	National Center for Education Statistics
CEEB	College Entrance Examination Board (school code system)
OPE	Office of Postsecondary Education (identifier system)
CRDC	Civil Rights Data Collection
SAIPE	Small Area Income and Poverty Estimates (U.S. Census Bureau)
F-33	Annual Survey of School System Finances (U.S. Census Bureau)
LEAID	Local Education Agency identifier (district-level join key)
AP	Advanced Placement (College Board)
IB / IBDP	International Baccalaureate / IB Diploma Programme
NAEP	National Assessment of Educational Progress
SEDA	Stanford Education Data Archive
GAO	U.S. Government Accountability Office
ISBE	Illinois State Board of Education
CPS	Chicago Public Schools
PCA	Principal Component Analysis
TF-IDF	Term Frequency–Inverse Document Frequency